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Changing Practices for an Innovative Care Pathway, the Mediating Role of the Coordinator

Zineb Cherkaoui^{a,b,c,d,e}, Caroline Merdinger-Rumpler^b, Patrick Pessaux^{a,c,d} and Célia Lemaire^{f,g,h}

^aDepartment of Visceral and Digestive Surgery, Nouvel Hôpital Civil, University Hospital of Strasbourg, Strasbourg, France; ^bEM Strasbourg Business School, HuManiS Research Center (UR 7308), University of Strasbourg, Strasbourg, France; ^cInstitute of Image-Guided Surgery (IHU Strasbourg), Strasbourg, France; ^dInserm Unit UMR_S1110, Institute for Research on Viral and Hepatic Diseases, Group « Relevance and Care Pathways », Strasbourg, France; ^eSSPC (Simplification of Surgical Patient Care), UR UPJV 7518, University of Picardie Jules Verne, Amiens, France; ^fMagellan Research Center (EA 3713), IAE Lyon, Université Jean Moulin Lyon 3, Lyon, France; ^gFaculté des sciences de l'administration, Laval University, Québec, Canada; ^hInstitut universitaire de France (IUF)

ABSTRACT

Getting professionals who have been selected and trained to perform individually to work together is a major challenge. The objective of the article is to identify the mechanisms by which a coordinator can ensure a mediation and boundary spanning role to enable the achievement of a project. Adopting a comprehensive approach, this article is based on the case of an innovative hospital project 'One-Day Diagnosis' (1DD) – one day instead of several weeks – of hepatobiliary and pancreatic pathologies, for which the professionals had to change their practices. Based on the results of the study, the findings show that the mediating role of the coordinator consists in making things happen and guiding health care professionals through the innovative pathway until it becomes a routine process. The study contributes to the literature on collaboration and relational coordination by highlighting that the change of practices for collaboration is compatible with personal excellence, by outlining the active role of the boundary spanner in the change of these practices and by demonstrating that the organizational stability of projects based on disembodied procedures is a necessary illusion that requires complementary interindividual relationships and a living leadership.

MAD statement

Extending the understanding of the ways to get individual performers to work together, the study highlighted how the change of health care professionals' practices is helped by the coordinator, a boundary spanner who makes things happen and accompanies the innovative pathway until its routinization. The study shows that collaboration is possible without giving up personal excellence and that the processual stability aim is an illusion which requires complementary interindividual relationships and a living leadership.

KEYWORDS

Coordinator; one-day diagnosis; care pathway; relational coordination; innovation

Introduction

The redefinition of the care pathway is something that is growing and in the future that will be a reorganization of our way of working [...] We will no longer practice medicine on the side without knowing what happened before and what will come after, coordination between the role players and all that. (PP1, quote from one of the interviews)

While some organizational innovations are important and interesting to implement for many stakeholders, the change in practices they impose can conflict with the professional culture of workers. This is even more prevalent for health care professionals (HCPs) who operate with a high degree of autonomy (Heldal, 2015), perform individually and in projects that impose cross-functional collaboration and efforts to change practices. Indeed, autonomy of health professionals and their need to be unique and incomparable are often considered incompatible with the conduct of collaborative projects, especially when medical but non-physician staff are involved and play a role in coordinating the project (Heldal, 2015). Particularly, the implementation of an innovative care pathway represents an organizational challenge that requires coordination mechanisms between HCPs to ensure the best care management to the patients as quickly as possible and to take into account the care already provided or to be provided by other HCPs of different specialties (Hernandez et al., 2013). The question that arises is to know what processes facilitate the coordination of innovative projects in such contexts also referred as boundary spanning in the literature. In health care contexts, the boundary spanning activities consist in integrating care across organizations, providers and settings, improving communications between care facilities and hospitals with the goal of improving patient care (Lewis et al., 2019). Although often associated with the boundaries between organizations, boundary spanning also occurs within organizations. Boundary spanners are then these individuals who play a communication role between the specialized units (Faraj & Yan, 2009) which units 'develop, each generates its own idiosyncratic norms, values, time frame and coding schemes to permit effective processing of information' (Tushman, 1977, p. 590). In this article, we focus particularly on the role of the project coordinator, seen as a boundary spanner to show to what extent her/his mediation role, already identified in the literature (e.g. Gittell, 2002a; Heldal, 2015; Lewis et al., 2019; Rundall et al., 2016), can have a real contributing impact in the launch and sustainability of the project. The objective of the article is therefore to identify the mechanisms that allow an actor to coordinate an innovative project that implies a concrete change of practice on the part of health care professionals. The article thus responds to the previously identified need to overcome barriers to improve the delivery, fidelity and evaluation of innovative projects by recommending facilitation strategies (Moussa et al., 2019), among which some have been proven to be effective, such as using goal setting, assessing progress and outcome, or providing tools and educational material, where the person who act as a facilitator has a crucial role for implementation (Moussa et al., 2019).

In this research, we adopted an abductive approach, seen as an inferential creative double-fitting data and theories process that enables a comprehension of facts (Dana & Dumez, 2015; Dumez, 2016; Peirce, 1935; Timmermans & Tavory, 2012). We investigated the role of the coordinator and explored how coordination mechanisms can be articulated during the implementation of an innovative patient care pathway: the 'One-Day

Diagnosis' (1DD) of hepatobiliary and pancreatic (HBP) pathologies. The 1DD is an innovative patient-centred approach which helps to reduce the duration to diagnosis and to the therapeutic decision from six weeks to only a day, while maintaining the requirements of diagnostic and therapeutic quality. This project is characterized by the six determinants of innovation in healthcare defined by Hernandez et al. (2013, p. 167): '(a) effective leadership, with the necessary technical and professional expertise and creative skills, (b) strong internal and external motivation to change, (c) clear and internally consistent organizational mission, (d) aligned organizational strategy, (e) robust organizational capability, and (f) continuous feedback and organizational learning.'

The findings confirm the importance of the coordinator when launching an innovative care pathway and highlight two essential roles. The first is to make things happen by relying on the patient-centred objective of the pathway to activate meaning and pride of the professionals involved in the project and by developing cross-functional communication throughout the pathway. The second shows that the routine implementation of these changes of professionals' practices is made possible by the recognition of the legitimacy of the coordinator and by the relational intelligence they develop during the project.

In this way, the study contributes to the literature on collaboration and relational coordination by highlighting the change dynamic of an innovative project in a health care organization. The findings show that when the characteristics of uniqueness and incomparability expected by professionals who function culturally in autonomy are transferred from an individual level to the level of a project, collaboration between professionals becomes possible and motivative. They outline also that the organizational stability of projects based on disembodied procedures is a necessary illusion to anchor practices, but it is not sufficient because it requires, in addition, continuous interindividual relations and a living leadership.

Theoretical Background

While collaborative approaches are beneficial to organizations, particularly healthcare organizations, they present significant challenges in their implementation. Working in multidisciplinary teams implies a change in the professional practices of each individual (Bruns, 2013), which underlines the inclusion in a learning dynamic, both collective and individual (Marsilio et al., 2017). Sharing the same goals is a source of motivation for HCPs to collaborate with others (Rundall et al., 2016). In particular, common goals help to connect participants and make it easier for them to understand the meaning of new information that arises. This contributes to the development of mutually acceptable responses that will allow the work of every HCP to contribute to the common goals (Rundall et al., 2016). Getting professionals to work together presupposes the implementation of specific managerial strategies, integrating in particular the existence of a shared objective and vision, but also leadership and favourable organizational conditions (Hernandez et al., 2013; Marsilio et al., 2017; Rundall et al., 2016). Exploring the role of the care pathway coordinator in this article, we drew on the literature on organizational coordination, particularly relational coordination, and on the coordination mechanism called boundary spanning.

In order to implement collaborative strategies in the field, organizational coordination, defined as the management of interdependencies between tasks, is required (Malone & Crowston, 1994). It has been shown that organizational coordination relies on formal and relational coordination mechanisms (Gittell, 2002a). Formal mechanisms are intended to have an impact on interactions among participants in a work process and it has been shown that routines, such as protocols or clinical pathways, team meetings, or boundary spanner positions, facilitate these interactions (Gittell, 2002a). Thus, coordination between professionals is based on practical modalities. Some authors point out the need to implement processes of operations management to allow adaptation and variability of procedures (Marsilio et al., 2017).

Relational coordination (Gittell, 2002a) is concerned with the modalities and logics at work during interactions between key players and refers to a more spontaneous form of coordination. Thus, in this study, the authors speak of mutual adjustment (Thompson, 2017) or teamwork (Van de Ven et al., 1976).

Gittell (2002a) and Gittell et al. (2010) proposed a conceptualization of relational coordination (RC) based on the role played by communication and relationships between professionals in the coordination process. It is defined as ‘a mutually reinforcing process of interaction between communication and relationships carried out for the purpose of task integration’ (2002b, p. 301). It has seven dimensions: ‘frequency, timeliness, accuracy, problem-solving nature of communication, relationships of shared goals, shared knowledge, and mutual respect’ (Gittell, 2002a, p. 1414).

Although work carried out in the health sector shows that RC is significantly associated with organizational performance, it plays a mediating role between formal coordination and performance (Gittell, 2002a; Gittell et al., 2010) and is intuitive in the empirical interactions between actors (Rundall et al., 2016). Although the actors are ‘aware’ of the coordination mechanism, it ‘is not yet a widespread management strategy’ (Rundall et al., 2016, p. 97). The study conducted by Rundall et al. (2016) shows that only four of the seven dimensions of RC are present in the minds of the accountable care organizations (ACO) leaders when they were interviewed and that these four dimensions are further reduced to two when they think of the mechanisms implemented at the operational level (frontline), particularly, for case managers. Thus, the frequency and problem-solving dimensions of communication are well integrated at both the leadership/executive and frontline levels, while the timeliness of communication and shared goals are only integrated at the executive level.

Effective coordination requires overcoming organizational boundaries. These boundaries are related to the diversity of functions, membership in separate departments, or the coexistence of multiple hierarchies, both formal and informal. Crossing organizational frontiers is encompassed through the concepts of boundary spanning and boundary spanners.

Boundary spanning refers to ‘activities that integrate the work of multiple people or departments’ in a given work setup (Lewis et al., 2019, p. 128). In a qualitative study conducted in ACO by Lewis et al. (2019, pp. 128–129), it was found that, boundary spanning activities can occur ‘across existing organizational boundaries, [...] provider specialties, [...] or across settings’. In our case, boundary spanning activities occur across existing organizational boundaries.

Boundary spanners refer to individuals whose main task is to integrate the work of others across boundaries of departments and specializations (Lawrence & Lorsch, 1967). In ACOs, for instance, ‘the role of boundary spanners is to work across multiple aspects of a patient’s care rather than focusing on direct care delivery in one setting’ (Lewis et al., 2019, p. 129).

In healthcare settings, boundary spanners include people like case managers, primary nurses (Gittell, 2002a), or pathway coordinators, as in our case, whose work is to understand the multiple aspects and dimensions of the patient’s care across organizational boundaries.

Previous research has shown that boundary spanners are increasingly effective under conditions of high interdependency of work processes and high uncertainty (Galbraith, 1973). These conditions apply to 1DD in which meeting the accuracy in time and quality of diagnosis is crucial.

In this article, we proposed that the concept of boundary spanning roles captures the notions conveyed by both boundary spanning activities and boundary spanners as individuals. Lewis et al. (2019) showed that new boundary spanning roles are developing as they study how boundary spanning activities are implemented in ACOs. They highlighted new boundary spanning roles through a new model of centralized coordination teams operating at the ACOs level. In our case, the boundary spanner is the care pathway coordinator of the 1DD.

Methods

The Research Context

The study was carried out at the Institut Hospitalo-Universitaire (IHU) in Strasbourg, France. The Strasbourg IHU is a healthcare and research center dedicated to image-guided and minimally invasive surgery, particularly digestive surgery. It is geographically adjacent to the Nouvel Hôpital Civil of Strasbourg University Hospital, one of its founding members. In addition to the implementation of cutting-edge surgical techniques, the IHU of Strasbourg is interested in care pathways as a whole, aiming to define, deploy and evaluate innovative pathways. Among these innovative pathways, the 1DD allows patients potentially carrying pancreatic, liver and/or biliary tract cancers or other hepatobiliary and pancreatic diseases to obtain a diagnosis with a treatment strategy in one day instead of the usual six weeks on average.

1DD Pathway: Process and Participants

The 1DD care pathway for HBP lesions is an innovative patient-centred care pathway set up by Strasbourg’s university hospital. Pessaix et al. (2018) described this new care arrangement with the objective of reducing the time required for diagnosis and the initiation of treatment, improving medical care for patients. The innovation consists in the new accelerated way of managing the patient pathway to diagnosing and delivering the appropriate treatment. Actually, the common care pathway includes multiple appointments with different HCPs, redundant complementary exams and imaging before establishing the right diagnosis and starting the treatment. The 1DD for HBP

diseases is an innovative care pathway that combines, on the same day, activities requiring usually several days of presence for the patients (detailed below) (Cherkaoui et al., 2023). The central role of imaging in HBP diseases allows in the majority of cases to make within the same day the diagnosis and the therapeutic decision while some situations need a histological confirmation after performing a biopsy particularly for malignant diseases.

The care pathway implemented includes the following features: making an appointment with a dedicated secretary, the first surgical consultation with the patient, imaging exams if required with the reservation of CT-scan and MRI schedules, a second surgical consultation to synthesize all new findings, anaesthesiology consultation if necessary, a therapeutic support consultation with a trained nurse performed for patients with cancer diagnosis, and, physical and psychological support to patients by a coordinator nurse who also coordinates HCPs (Pessaux et al., 2018).

Furthermore, a dietician and psychologist will join the 1DD working staff to offer additional specialized consultations as soon as financial resources are available for recruitment procedures.

All the steps of the 1DD are run in the same place, at the Institute of Image-Guided Surgery (IHU of Strasbourg), which encompasses the consultation rooms, operating rooms and radiology department. Thus, the complete patient's care pathway takes place in the same location.

The accelerated patient management for HBP lesions took place twice a week on Tuesday and Wednesday, with dedicated sessions for each of the two hepatobiliary and pancreatic surgeons. Knowing the existence of this consultation model makes general practitioners and other medical specialties professionals directly addressing their patients to join this care pathway avoiding wasting time performing inadequate investigations.

The Pathway Coordinator: A Central Position

The role of the coordinator throughout the 1DD pathway is central for the patient as well as for the HCPs. Providing psychological support at each stage and guiding and accompanying the patient, she is a reference person at each stage of the care pathway. For HCPs, she is the person who controls the entire process and coordinates everyone's actions, thus playing a pivotal role for the 1DD, notably as a boundary spanner. Indeed, the implementation of an organizational innovation of accelerated care pathways, such as the 1DD, requires a single location for each of the diagnostic stages. In addition to increased comfort for the patient, the physical proximity of those linked to the pathway is essential. The proximity of the professional with the patients in terms of location is a key factor that affects the emergence of quality interactions between them. For this type of organization, temporal coordination implies establishing dedicated 1DD slots for each of the professionals to meet the challenge of the promised accelerated diagnostic and therapeutic management.

The Research Process

In this research, we adopt an epistemological approach inspired by the Peircian conception of abduction (Peirce, 1935), seen as the reasoning by which one seeks to

explain a fact or a set of surprising facts, and which makes it possible to produce new ideas through the best explanation of the reality. Aiming at identifying the mechanisms that allow an actor to coordinate an innovative project that implies a concrete change of practice on the part of health care professionals, we rely on 'knowledgeable agents' feedback to better understand the process (Dana & Dumez, 2015, p. 157).

The study that we conducted is based on semi-structured interviews among HCPs who participate directly in the 1DD care pathway or work closely with it. We interviewed 19 HCPs between April and July 2019, knowing that the innovative organization was set up in January 2017. We contacted each HCP in person or by e-mail. The participants received information on the relevance and design of the study, the problem, the central concepts of the study and the way the results would be distributed. Each participant explicitly consented to participate in the study. The interviews were conducted individually by the same interviewer, following the same outline. The interview guide incorporated following subjects. First, we focused on the timing and circumstances during integrating the 1DD care organization. Subsequently, we discussed the proper experiences, expectations and constraints in the interviewees' daily work in 1DD. We then covered 1DD success key factors, threatening elements and recommendations, highlighted according to each respondent. Finally, the HCPs described their vision of the future of the 1DD care organization.

Study Sample

Our study sample was composed of 19 HCPs subdivided into 10 categories. Table 1 presents an overview of all the interviewed persons. In order to capture the diversity of professional experiences, we interviewed both pioneers engaged since the early days of the project and HCPs involved later in the 1DD for each category. We interviewed the two hepatobiliary and pancreatic surgeons who performed specialized medical consultations and two radiologists who worked in pairs with these surgeons. Ten anaesthetists who work in the IHU operating room could be requested to make a consultation on the same day for patients with oncoming surgery. Two interviews were conducted among this group of ten anaesthetists. One was a pioneer and the other one had joined the 1DD after its implementation. We conducted an interview with the 1DD hepatobiliary and pancreatic dedicated medical secretary, the two welcoming hostesses and the two therapeutic support nurses involved in the accelerated pathway of patients with diagnosed cancer. Among the radiology staff, we interviewed the representatives of radiology operators, one out of the three current secretaries, and the first secretary involved in the 1DD care organization at the beginning of the project, who did not work in the care organization anymore. We also interviewed the two persons dedicated to this project regarding its implementation: the programme director and the coordination nurse. In addition, the head of the IHU operating room, the future dietician and a medical secretary working in the open space next to her college involved in this organization, were also interviewed.

Data Collection and Analysis

The interviews were recorded with the consent of each HCP, fully transcribed and anonymized. We assigned codes to each one (Table 1). We analyzed the qualitative data

Table 1. Characteristics of the one-day diagnosis healthcare professionals.

HealthCare Professionals (HCP) Category code	HCP Category	HCP	Involved actors in One-Day Diagnosis pathway	Number of interviewed actors N (%)	Interviews code
1	Surgeons	HepatoBiliary and Pancreatic (HBP) Surgeon	2	2 (100%)	FE1; PP1
2	Radiologists	Radiologist	2	2 (100%)	FV2; HS2
3	Anesthetists	Anesthetist	10	2 (20%)	MJ3; PA3
4	Secretaries	1DD medical secretary	1	1 (100%)	BS4
4	Secretaries	Medical secretary	not directly involved in the 1DD	1	KC4
5	Reception desk	Welcoming host	2	2 (100%)	CJ5; SJ5
6	Nurses	Therapeutic support nurse	2	2 (100%)	HN6; HS6
7	Managers	Head of operating room	not directly involved in the 1DD	1	MR7
8	Dieticians	Dietician	not involved in the 1DD yet	1	ME8
9	Radiology staff	Representative of radiology operators	NA	1	BM9
9	Radiology staff	Radiology current secretary	3	1 (33%)	KE9
9	Radiology staff	Radiology former secretary	not involved in the 1DD anymore	1	KV9
10	Organizational team	Nurse coordination	1	1(100%)	LE10
10	Organizational team	Programme director	1	1(100%)	LC10
TOTAL			25	19	

collected following an abductive approach and performed a two-step coding process involving an interplay between data and theories to light on understanding the observed phenomenon (Dana & Dumez, 2015; Dumez, 2016; Timmermans & Tavory, 2012).

First, we proceeded with open coding for each interview and then by topic covered in the interview guide to prevent from the risk of circularity. Among the numerous first order themes that emerged, we chose to refine our research focus on the role and contribution of the coordinator. The second coding consisted in 'back and forth confrontations' (Dana & Dumez, 2015, p. 160) between data regarding the role of the coordinator in the 1DD care pathway and theory based upon the framework of organizational coordination (Gittell, 2002a, 2002b). The three main themes that emerged from these successive stages are linked to the roles of the coordinator in the 1DD process:

- The first theme is linked to the coordinator acting as a boundary spanner that's to say integrating the work of people engaged in an interdependent system,
- the second theme is linked to technical, processual, organizational coordination in a formal way,
- the third theme is linked to the transitional role of the coordinator of the 1DD programme, in the sense that the coordinator is not thought as a permanent function, but as limited in time.

All the interviews were analyzed again, and meanings units were coded following those three themes, allowing us to say that in a spatiotemporal context, it can be said that the coordinator is a border-crosser, advising professionals, guiding and linking them, and has a role in the perpetuation of the accelerated diagnosis model.

Results

The study shows that the role of the coordinator of an innovative care pathway is organized around two central concerns. The first is to think and lead the innovative process and the second consists of preparing for the future by making the process evolve so that the pathway is 'put into orbit' without further supervision of the coordinator.

The Pathway Coordinator Makes Things Happen

This innovative pathway is a challenge to implement since it is more demanding for the HCPs involved compared to their usual activity (constraints on dedicated slots for radiologists and surgeons, overload for anaesthetists with additional appointments ...) and thus can be seen as negative by HCPs if it is perceived as a constraint on their autonomy (Heldal, 2015). As an interviewee states: 'Implementation is a daily struggle' (PP1). At the same time, the participants of the innovative pathway are fully involved, as a representative of radiology operators explains to us:

Everybody was playing the game, that's it. It's the only way to do well in the end. Everybody (was) in tune (BM9).

In the case of 1DD, the coordinator is the key player who makes it possible to move from the ideation stage of a care pathway to its effective implementation. This change process is effective thanks to the mechanisms detailed in the following paragraphs.

Relying on the Patient-Centred Objective of the Pathway Activates Meaning and Pride of the Professionals

All HCPs shared a federative objective based on patient benefit. The idea of increasing opportunities and reducing overall patient stress was the starting point for the launch of the 1DD and motivated the commitment of the HCPs in this innovative approach. With regard to relational mechanisms (Gittell, 2002a; Gittell et al., 2010), within the framework of the 1DD, the pathway coordinator supports the construction of strong interprofessional relations built around the awareness of the perceived benefits for the patient. She/he indeed included the patients in the improvement process, considering their needs and feedback.

I followed patients closely, especially at the beginning, when they had just been diagnosed with cancer (...) it is interesting to have both the healthcare professional/the medical technical perspective, and also the feedback from the patient (LE10).

The motivation of the professionals lies in the fact that the pathway is patient-centred. Some HCPs feel a certain pride in contributing to a 'pioneering journey' (KV9) or are stimulated by the challenge that this innovative journey represents for them. Others express an increased interest in the opportunity to participate in a 'beautiful experience' (BM9) or in

the development of a more stimulating professional practice. Professionals express their pride in contributing to this organizational innovation.

The patient's care is objectively faster. The general practitioners are all enthusiastic, frequently I call them to communicate, to inform them etc. or sometimes I call them for other patients and tell them that we are doing this ... everyone is happy because it is one of the problems, all over the world, the waiting times, especially in the hepatobiliary field where you have to redo a scan, an MRI, an anaesthesia consultation ... it really saves a month ... for a pancreatic cancer it is important. (FE1)

In addition to the shared objectives, which are important for relational coordination, this contributory pride is a supplementary catalyst of motivation specific to innovative pathways. 1DD participants, practitioners find meaning and improve identification with the organization (Specht et al., 2018).

Developing Cross-Functional Communication to Fluidify the Care Pathway

The communication mechanisms between professionals are a strong point of the 1DD.

I would say it's a good circuit when everything is running smoothly, everyone is getting along, and everyone is helping each other and everyone is willing to go out of their way to make sure the patient gets everything in one day and the surgery date comes as soon as possible (MJ3).

The HCPs involved at the time of the launch of the pathway developed informal communication logic among themselves, enabling them to exchange information useful to the achievement of this new pathway and to build new cross-functional professional practices.

Communication is good with the surgeon, it is true that since we see the files together and we discuss them, there is a consensus and we manage to do something that is correct from his point of view and from mine I think, so I think that communication is certainly the most important part in judging what to do with the patients and that is well done (FV2).

The pathway coordinator supports the communication mechanisms implemented by 1DD professionals. Ensuring that the frequency of interactions is maintained between them and relying on their availability, the coordinator promotes exchanges aimed at problem-solving (Rundall et al., 2016), such as the arrangement of the slots of the various HCPs and their articulation during the day of diagnosis.

When the therapeutic support nurse needs information on the pathology and does not know enough about it, she will call the surgeon to ask for more information and to be able to carry out the consultation [...] afterwards, it is also the role of the coordinator to check that things are working well, to check that the right information is given to the right people at the right time and in the right place, and so, all of this ensures that things work well (LE10).

The coordinator ensures that the mechanisms developed during the early days are maintained and transformed in a constructive manner between professionals participating in the 1DD, especially when new professionals are recruited.

The messages are getting through to people, we always manage to inform each other (HS6).

The coordinator was often present and explained the process to us (SJ5).

Favouring the development of organizational coordination upstream of the innovative care pathway project, the coordinator plays the role of a boundary spanner between the professionals involved in the 1DD when the links between them do not occur spontaneously. Facilitating exchanges, the pathway coordinator supports relational coordination in its two dimensions of relationships and of communications between the key individuals, which has been shown as a condition for organizations' performance (Gittell, 2002a; Rundall et al., 2016). Indeed, acquisition of communication skills is essential (Rundall et al., 2016) which helps create a virtuous circle of innovation specific to multi-disciplinary teams (Marsilio et al., 2017). For example, by organizing pathway meetings, the coordinator ensures that informal elements of RC are combined with formal elements of the clinical pathways.

In order to establish the legitimacy of the project and to ensure a distributed leadership among several individuals in the process, the implementation of communication that goes beyond the framework of the existing team is, according to some of the interviewees, desirable. This communication would disseminate the practice of the innovative pathway to colleagues, patients, practitioners and authorities. It would make it possible to accentuate an existing collective dynamic of pride of belonging that feeds the commitment of the role players in the 1DD pathway.

The Coordinator Accompanies the Innovative Pathway Until it is put into Orbit

Initially, the action of the coordinator is indispensable for the 'launch' of the innovation pathway, at which point, his role is personified as the boundary spanner, whose responsibility is to integrate the work of professionals from different services, functions and hierarchies (Gittell, 2002a; Rundall et al., 2016). Subsequently, for the 'putting into orbit,' i.e. to move from 'bricolage' (LC10) to formalized processes; the coordinator must step aside to make way for the coordination function (boundary spanning), which must be shared between the team members participating in the care pathway. The leadership of the project must be shared in order to ensure the transformation of this role into real organizational capabilities, which are crucial for innovation (Hernandez et al., 2013). The role of the coordinator is essentially to oversee that the key players have sufficiently appropriated the tasks and roles, that also involves boundary spanning roles, so that there is no further requirement of his individual role in the pathway. In other words, she/he must organize the integration of his or her own role into a routine process for all HCPs involved in the 1DD.

The replicability of the model depends on the flexibility of the organizational, human and financial resources of the structure. The valorization of the acts must, especially, encourage innovative activities (MJ3), but beyond that, the project must be carried by several professionals who accept to change their practices and who develop their relational abilities.

Giving Trust and Legitimacy to the Coordinator to Integrate the Changes into Practices

The practices of the professionals change with the implementation of the 1DD. With a usual organization, a surgeon can schedule ten consultations per half-day, whereas with the 1DD, seeing these same patients can take the whole day. In addition, she/he

walks much more than usual in the hospital to exchange with different colleagues. In particular, she/he will consult with the 1DD radiologist based on imaging results because she/he feels that a simple report is not sufficient for interpretation. In the same way, the work flow of anaesthetists is changed with the 1DD, as an anaesthetist states:

This is extra work for anaesthetists who have to leave the operating room to provide a consultation, whereas normally consultations are grouped together. This is extra work for us, and the operating room are running without an anaesthetist for the time of the consultation (MJ3).

In the innovative organization of the 1DD, time management is thus crucial. The unavailability of a professional or an unspecified niche creates a significant risk of 'missing' the objective of the 1DD.

What could threaten the 1DD? I think it's the availability of radiologists for example, I'm going on vacation, who will take over on Wednesdays? (HS2).

As a consequence, the 1DD requires a fluid articulation of all the specialist consultations necessary to establish a diagnosis with a resulting therapy and a plan of care delivered to the patient. This formalization involves the inclusion of planning in a routine. The interviewees insist on the importance of temporal organization: the availability of actors in slots dedicated to 1DD is a condition for the success of the accelerated course. This implies a preparation upstream of this process, which may seem like a 'waste' of time and a complication of the work but is in fact an essential phase in the smooth running of the 1DD. Radiologists, stressing this point, indicated the importance of dedicating time for this journey. The lack of time leads to the non-execution of 1DD on a given day.

Beyond the planning that changes professionals' practices, the nature of the one-day diagnosis also changes the content of their work. In particular, the rapidity of the announcement modifies patient behaviour and imposes a change in practices for professionals. The absence of a delay between the examinations and the diagnosis, indeed, changes the nature of the questions asked by patients. Less distracted by the post-treatment period and potential complications, they asked professionals more about the pathology itself.

The patients have stored up a lot of information on the same day, so they don't necessarily have the same questions on the day of the diagnosis when they have had all the information, the anaesthetist and everything, they have plenty to think about. And a little bit at a distance where there are other questions that may come up and they may have other thoughts (HS6).

These technical questions required some of them to update their knowledge. In short, the announcement of the diagnosis to the patient, always a key moment during the treatment, represents an important change of the work that must be the subject of specific training.

The coordinator also participates to the perennity of the process when she/he replicates the HCPs' cross-functional professional practices developed in this innovative pathway. She/he plays the role of relational interface, when, for example, she/he institutionalizes a new practice of common diagnosis during a real-time dialog of the patient's case between a surgeon and a radiologist. While the design and implementation of this common diagnostic practice was based on the existence of a shared willingness by the pioneering surgeon/radiologist pair, the replication of this practice by a second pair

was facilitated by the support of the coordinator. Based on the model built by the pioneering pair (Surgeon 1 and Radiologist 1), the coordinator accompanied two professionals who have no knowledge of each other (Surgeon 2 and Radiologist 2) to build a relationship around the establishment of a common goal, shared knowledge and mutual respect between the professionals (Gittell, 2002b). This combination accompanies and anchors the construction of new organizational routines (Rundall et al., 2016), such as the co-production of a common and simultaneous diagnosis between surgeons and radiologists.

In this changing process, the coordinator must explain the role of each person in the pathway so that they have an overall view and she/he must ensure the existence of favourable organizational conditions. An essential condition to ensure the success of these important changes is that professionals give legitimacy to the coordinator. This process is possible because the physicians placed their trust into a professional who doesn't belong to their professional category. Formal leaders must agree to give up some authority and control in favour of the coordinator who provides a leadership without hierarchical or symbolic authority (Uhl-Bien et al., 2007). The 1DD succeeded because of the recognizing of the leadership of the non-physician pathway coordinator.

The Relational Intelligence of the Individuals, a key for Achievement

The success of the project relies on the one hand on strong leadership, first embodied and then distributed among several people, and on the other hand on interpersonal relationships that act as a cement between processes that are by nature disembodied.

To sustain the project, the existence of real leadership goes hand in hand with the need for strong will. The 1DD is led by a strong leader, a surgeon specialized and legitimate (PP1). The programme director of the 1DD expresses this: 'having a hospital project leader who is mobilized and a little bit the flagship resource and who is present for the others, it helped' (LE10). According to the leader himself, the project must be supported by someone who is convinced of the project's benefits:

The key to success is that there is someone who really believes in the project and can regularly give reminders to defend it (PP1).

Going beyond this embodied dimension of leadership implies ensuring the constitution of a 'chain of expertise' (LC10) based on professionals involved in the process, trained, and aware of their interdependencies. In the 1DD project, this chain of expertise began with the coordinator, who benefited from the legitimacy of the leader, in the sense that the leader trusted the coordinator, and thus, the other participants recognized the legitimacy transferred to the coordinator. After setting up the project, the coordinator prepared her own succession, distributing the roles she assumed and thus ensuring the rise in competence of the multidisciplinary team (Marsilio et al., 2017).

In particular, the surgeon's secretary assumed an essential role in organizing the schedules in the one-day diagnosis. Thus, the participants in the 1DD all gave their trust and legitimacy to personnel of different status and who evolved over time. This distribution of leadership is a condition for the success of the project. Beyond the success of the project, the recognition of the expertise of people who are not usually involved contributes to their well-being at work: they are valued because they bring value to the project and

their opinion counts. Indeed, the process rely on people, who are the real actors in the process.

My experience is positive, I'm happy to participate as a secretary, to be able to share my ideas, to be asked for my point of view, I think it's good to involve the bottom of the ladder (BS4).

The success of the 1DD rely also on the quality of the relationships between individuals: the relational skills of the actors are central to the sustainability of the system. To stay at the level of quality we have now, we also rely on the qualities of relationships, empathy, etc. The system also relies on people, and I think that's why it works well (LC10).

I think that the well-working relationship between the surgeon and the radiologist is remarkable. I think it's exemplary, and it is one of the key points in the success of this pathway (BM9).

The analysis over time is better on Tuesdays because it happens since the beginning, whereas Wednesdays are recently concerned by 1DD. So, there are mechanisms to put in place, necessarily. You change individuals, you change ... Even if the organization remains the same, from the moment you change individuals, they are the ones who finally do ... most of the work (BM9).

The pathway in this study is based on the voluntary work of people who are involved and who maintain good relationships. The dynamics of change therefore rely on people who are personally committed to the project, who make efforts to nurture also their inter-personal relationships.

Here, each one contributed by giving themselves and tried to see what was possible to do to find together; what was the best possible organization to make it work for everyone (LE10).

Sometimes it's out of the slots, you have to negotiate an anaesthesia consultation, an imaging, so it depends on the radiologist, the anaesthesiologist who are more or less ... For example, the other day I asked for an anaesthesia consultation very, very late, there weren't many patients in the OR, the patient lived 200 km away, she made the effort ... (FE1).

Thus, the sustainability of the project depends not only on the routinization of practices or their formalization in procedures, but also on the intention of individuals to participate by collaborating, which requires the ability and desire to build or maintain quality interpersonal relationships.

Discussion

The Change of Professional Practices Through Collaboration, Without Giving up Personal Excellence

The medical population is reputed to be reluctant to be managed (Heldal, 2015), especially to develop teamwork (Boyle & Kochinda, 2004). Indeed, since the selection process during medical studies, individuals are in competition, they must demonstrate individual excellence, but relational dimensions are never valued, despite their known value (Stecher & Ketcham, 2022). However, as the results of our study show, the success of 1DD's innovative project is conditioned by the development of relational capacities at several levels. When the leader transmits his legitimacy to the coordinator, he accepts to let go some of his personal power for the benefit of another person and then for the benefit of a collective, while keeping the responsibility of the project (Uhl-Bien et al., 2007). When individuals make efforts to preserve their interpersonal

relationships or name the patient's interest, they also show relational skills that challenge the idea that physicians aim to maintain their unique and incomparable character (Heldal, 2015). They are willing to change their practices in the name of a project that seems important to them (smoothing pathways for patients), to achieve a desired outcome (Reay & Hinings, 2009). The uniqueness and incomparability expected of professionals is in some ways transferred or projected onto the 1DD project. By participating in the project, they are no longer unique and incomparable for the medical feats they perform, but because they agree to collaborate on a project that is itself unique and incomparable. Two mechanisms activate the internal motivation of professionals. On the one hand, the project feeds their desire to be unique and incomparable by transcending the individual dimension for the benefit of the collective project. On the other hand, the collaboration leads some professionals to work 'with' the other, simultaneously, like the surgeon–radiologist pair who interpret images together. The learning mechanisms that are put in place are motivating for these professionals who can develop new skills. By activating micro-processes specific to the functioning of professionals (Reay & Hinings, 2009), the 1DD encourages their commitment to a project to the point of changing their practices, which helps to reinforce their initial commitment. It is therefore important, notably for the boundary spanners, to preserve the integrity of the professionals' motivational factors to ensure the success of the project. This study shows indeed that boundary spanners not only have a communicative role, as many studies suggest, but that they also intervene in the profound change of practices. They are high-impact change agents. When practitioners evolve their practices, they also shift the way they see their own profession by including the collaborative dimension, thanks to the intervention of the boundary spanner.

From the Necessary Illusion of Stability to a Living Leadership, Leaving Room for Interindividuality

The challenge of moving from the individual role of the boundary spanner held by the coordinator to a collective role of anchoring the cooperative practices necessary to achieve the innovative course is central to this study. Our research meets the findings of Lewis et al. (2019) and sheds light on the different modalities that the role of boundary spanner can take in an organization or project, from the individual to the team. The setting in orbit that we have described is an initial intention for which the coordinator has developed organizational mechanisms that promote the stability of the processes to ensure the sustainability of the project. However, the results show that the anchoring in practices by professionals responds not only to the application of existing procedures but also to the intention of nurturing interpersonal relationships that structure the modalities of involvement in the project. The stability of the processes is therefore an illusion because the mechanisms live and work if the professionals agree to carry them out. The recognition of the coordinator's legitimacy depends on the fact that the professionals recognize the legitimacy of the project leader. Therefore, the interpersonal relationship is central and avoid misunderstandings and disputes that can occur when non-medical coordinators are involved in a project (Heldal, 2015). Similarly, when professionals make themselves available so that the patient does not have to multiply trips for imaging and other appointments before the start of treatment and/or surgery, they

make efforts in the name of the patient's interest and the relationship they have with their colleagues. Thus, the 1DD, like many projects, responds to a logic of accountability to colleagues to nurture interpersonal relationships at work. This characteristic can be considered a priori as a fragility of the process. The system is fragile because it relies on the acceptance of professionals to commit and remain involved, with a particular challenge appearing in the retention and attractiveness of new doctors to ensure the reliability of the sufficient number of people contributing to the project. Moreover, interpersonal relationships are not the desirable way for organizations to function. However, to imagine that they do not exist seems to us to be an error of appreciation of existing mechanisms. Rather than deploring the existence of these mechanisms, it would seem interesting to us to rely on these characteristics to build a living leadership mode, particularly adapted to the complexity of certain contexts (Pepper, 2002). It is then a matter of considering the importance of nurturing interindividual relationships to build new organizational modalities together. Thus, by relying on arrangements at micro levels (Reay & Hinings, 2009), professionals maintain their independence while working together. This idea is similar to the one according to which self-organization is desirable in the medical world (Mintzberg, 2018), and opens up the modalities of collaboration to different worlds (those of care, cure, control and community). A point of vigilance should be noted concerning this interpersonal dimension: developing a living leadership that integrates the importance of interpersonal relations requires the development of an ethical awareness on the part of professionals in order to avoid the appropriation of the project for the benefit of individuals, groups of individuals, alliances or dominant positions.

Conclusion

The study detailed the roles of the coordinator of an accelerated care pathway, who acts as a boundary spanner between professionals and develops transitional mechanisms to ensure the sustainability of the care pathway. Firstly, when the care pathway coordinator makes things happen, her/his role is based on communicating the patient-centred objective of this pathway, supporting the commitment of each of the actors and reinforcing their professional interest on a regular basis. Our findings show that this process relied on the HCPs' convictions of the interest of the project, that nourished their meaning and pride of participating. Moreover, at this stage, the coordinator develops cross-functional communication making the pathway more fluid while supporting existing mechanisms of communication between actors. Secondly, the coordinator accompanies the innovative pathway to reach its sustainability until it becomes a routine process (put into orbit). At this stage, she/he could support the change of the practices of the professionals since she/he was considered as legitimate by them. In addition, the sustainability of the process relies on the relational intelligence of the professionals.

The research identifies three major contributions to the literature. First, regarding the collaboration of professions that culturally operate in autonomy, such as physicians, we show that the need to combine uniqueness and incomparability specific to these professions (Heldal, 2015) can be compatible with collaboration with other professionals if the project being conducted meets the same characteristics of uniqueness and incomparability. Thus, if physicians' expectations are transferred from the individual level to the

project level, which then endorses these expected characteristics. Secondly, while previous literature states the importance of communicating role of the boundary spanners (e.g. Lewis et al., 2019), our study expands it by showing the active role the boundary spanner has to trigger the change of the professionals' practices. Thirdly, we contribute to the literature on relational coordination (Gittell, 2002a; Hernandez et al., 2013; Rundall et al., 2016) by showing that the illusion of organizational stability based on disembodied procedures is necessary to temporarily stabilize practices in projects that must, however, rely on interpersonal relationships and a living leadership to endure.

Our research has two important limitations. The study is mono-centric and is based on a unique case of an innovative pathway. The prospect of studying the role of the coordinator and the mechanisms implemented in other innovative paths, if possible, in a longitudinal approach, appears promising. In future research, it would be relevant to study how the coordination mechanisms are set up and whether the virtuous circle of coordination is maintained over time. Identifying how professionals continue to learn and change their practices appears to be an essential managerial challenge for the sustainability of this type of programme. In particular, when new members join a multidisciplinary team, it would be wise to study the handover to understand how good relational communication in the sense of Gittell (2002b) is maintained with the integration of new members. In addition, our results suggest a research perspective that could consist of the specific study of the pride in contributing to an innovative pathway, in order to identify the associated motivational mechanisms in healthcare teams. Coordination is a way of leading professionals and thus is both a science, which imposes a rigorous method, responds to a way of thinking about work (a philosophy) and a way of working (Pepper, 2002).

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Institutional Review Board Statement

The study was conducted in accordance with the Declaration of Helsinki. Ethical review and approval were waived for this study as it is routine.

Informed Consent Statement

Informed consent was obtained from all subjects involved in the study.

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